Physics of Fluids

Selfsimilar expansion of dense matter due to heat transfer by nonlinear conduction

R. Pakula and R. Sigel

Citation: Phys. Fluids 28, 232 (1985); doi: 10.1063/1.865184
View online: http://dx.doi.org/10.1063/1.865184
View Table of Contents: http://pof.aip.org/resource/1/PFLDAS/v28/i1
Published by the American Institute of Physics.

Self-similar expansion of dense matter due to heat transfer by nonlinear conduction

R. Pakula and R. Sigel
Max-Planck-Institut für Quantenoptik, D-8046 Garching, Federal Republic of Germany

(Received 30 May 1984; accepted 4 September 1984)

The hydrodynamic flow, called the ablative heat wave, from a dense body which is suddenly brought into contact with a thermal bath with a temperature of the order of 10^5–10^8°K is investigated theoretically. It is shown that the hydrodynamic equations with nonlinear conduction admit a self-similar solution of the asymptotic type, i.e., approached by the flow for $t \to \infty$. Self-similarity is a consequence of the fact that the solid may be considered as infinitely dense in this limit, even for a bath temperature varying in time according to $T \sim t^\tau$ (within certain limits for τ). The self-similar solution is calculated in detail, and scaling relations for the hydrodynamic variables are given. Numerical values are obtained for the radiative heating of gold. The range of validity of the solution is discussed.

I. INTRODUCTION

A. Statement of the problem

Consider a situation where the one half-space is filled by a body of solid density; the other half-space is empty [Fig. 1(a)]. At time $t = 0$ the body is brought into contact with a thermal bath with a temperature of the order 10^5 to 10^8 °K. Because of heating, a steep temperature gradient is set up, leading to the diffusion of heat into the interior of the body. Simultaneously, a high pressure is generated with the consequence that hydrodynamic expansion starts at the surface of the body. As time goes on, the heat supplied by the bath is continuously converted into internal and kinetic energy of the heated material. Our interest is to calculate the net heat flux from the bath to the material as a function of time. Because heating and expansion occur simultaneously, the solution of that problem can only be found by solving the hydrodynamic equations in space and time, taking heat conduction into account.

This problem is in fact a classical problem of the physics of hydrodynamic phenomena; as a general reference we note the book of Zel'dovich and Raizer.[1] As described there (see Ref. 1, Vol. II, Chap. X, Sec. 8), the heating of the body for a constant bath temperature proceeds as follows: first a supersonic heat wave propagates into the undisturbed material of solid density ρ_s [Fig. 1(b)]. As the thickness of the heated material increases, the velocity of the wave decreases. At the same time, hydrodynamic expansion of the heated material into the vacuum begins. When the front of the heat wave has slowed down to approximately sound velocity, it is overtaken by a shock wave [Fig. 1(c)]. The subsequent state where the heat wave is preceded by a shock wave and accompanied by hydrodynamic expansion, is maintained for later times. We call it the ablative heat wave.

In our investigation we concentrate on the ablative heat wave with the aim of finding a self-similar solution of the asymptotic type, i.e., a solution which the actual flow approaches in the limit $t \to \infty$. Obviously, such solutions are of particular interest.

Our treatment is limited to the so-called conduction approximation. In this approximation the physical nature of the bath and its interaction with the material are left out of consideration; only the bath temperature appears as a boundary condition for the first Lagrangian cell of the body. As a conduction mechanism in the material, we deal mainly with radiation heat conduction in a multiply ionized material. It should be emphasized, however, that the analysis is performed for an arbitrary power dependence of the coefficient of heat conductivity on density and temperature and also applies to, in particular, the case of electronic heat conduction. The range of validity of the conduction approximation will be discussed at the end of the paper (in Secs. V and VI).

B. Hydrodynamic equations with nonlinear heat conduction

At the envisaged temperatures of the order of 10^5 to 10^8 °K, the coefficient of heat conductivity κ depends on the

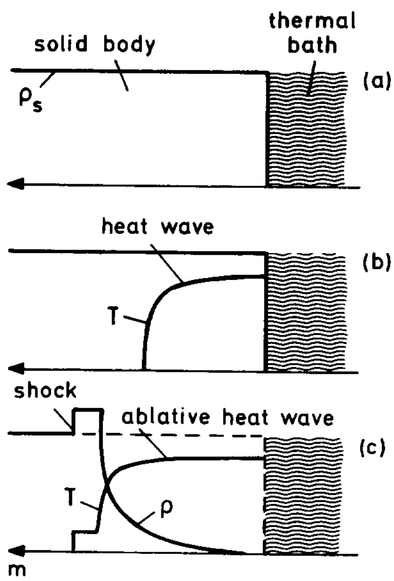

FIG. 1. Heating of a solid body in contact with a thermal bath: (a) at time $t = 0$ the body is brought into contact with the thermal bath; (b) for $t > 0$, first a nonlinear heat wave runs into the undisturbed material; (c) subsequently, hydrodynamic motion of the heated material becomes important, and the heat wave is overtaken by a shock wave and the ablative heat wave forms.

 0031-9171/85/010232-13$01.90

temperature and density of the material. It is usually assumed to have a power dependence on these quantities. That is, with S the heat flux, T the temperature, and $v = 1/\rho$ the specific volume (ρ the density) we have

$$S = -\kappa \frac{\partial T}{\partial x}; \quad \kappa = \kappa_0 T^{\nu} v^{\mu}.$$

The coefficient κ_0 and the exponents ν and μ depend on the conduction mechanism.[1] With radiation heat conduction one has $\nu \cong 4$–6, $\mu \cong 1$–2; with electron heat conduction and a fully ionized plasma $\nu = 5/2$, $\mu = 0$.

In planar geometry, the hydrodynamic equations including nonlinear heat conduction, but not radiation pressure and energy, are[1]

$$\frac{\partial v}{\partial t} - \frac{\partial u}{\partial m} = 0, \tag{1}$$

$$\frac{\partial u}{\partial t} + \frac{\partial p}{\partial m} = 0, \tag{2}$$

$$\frac{1}{\gamma - 1} \frac{\partial (pv)}{\partial t} + p \frac{\partial v}{\partial t} = a \frac{\partial}{\partial m}\left((pv)^{\lambda} v^{\mu - 1} \frac{\partial (pv)}{\partial m}\right)\left(= -\frac{\partial S}{\partial m}\right). \tag{3}$$

Here m is the Lagrangian mass coordinate, u, v, p are, respectively, the velocity, specific volume, and pressure, and γ the adiabatic index. To put the hydrodynamic equations into the form given above, we have taken the specific internal energy e as

$$e = pv/(\gamma - 1), \tag{4}$$

and the equation of state as

$$pv = KT^{\delta}. \tag{5}$$

The constant a is related to κ_0, K, ν, and δ by

$$a = \kappa_0/(\delta K^{(\nu + 1)/\delta}). \tag{6}$$

The exponent λ is related to ν and δ by

$$\lambda = (\nu + 1 - \delta)/\delta. \tag{7}$$

C. Self-similarity

In this paper we wish to derive a self-similar solution of the problem stated above. In order to contrast clearly our results to previous ones, we find it necessary to start with a brief outline of the concept of self-similarity. On the basis of Buckingham's theorem,[2] it has been discussed, together with many applications, in the books of Sedov[3] and Barenblatt.[4] We basically follow the notation of Barenblatt.[4]

Let a physical quantity g depend on n dimensional parameters $a_1,\dots,a_n$;

$$g = g(a_1,\dots,a_k,a_{k+1},\dots,a_n),$$

where g is called a governed quantity, the parameters $a_1,\dots,a_n$ governing parameters. The latter have been grouped so that $a_1,\dots,a_k$ have independent dimensions (otherwise the choice of $a_1,\dots,a_k$ from $a_1,\dots,a_n$ is arbitrary). It is assumed that the dimensions of the parameters $a_{k+1},\dots,a_n$ as well as of g are expressible in terms of power monomials of $a_1,\dots,a_k$. Then, according to the Buckingham theorem, g can be expressed as a product of a dimensional function g_1 and a dimensionless function $\widetilde{G}$, where $\widetilde{G}$ depends on n-k dimensionless similarity parameters $\xi_1,\dots,\xi_{n-k}$,

$$g = g_1(a_1,\dots,a_k)\widetilde{G}(\xi_1,\dots,\xi_{n-k}),$$

and g_1 is of the form

$$g_1(a_1,\dots,a_k) = a_1^{\alpha_1} \cdots a_k^{\alpha_k}.$$

The dimensionless similarity parameters are

$$\xi_1 = a_{k+1} \cdot a_1^{\alpha_1^1} \cdots a_k^{\alpha_k^1}, \dots, \xi_{n-k} = a_n \cdot a_1^{\alpha_1^{n-k}} \cdots a_k^{\alpha_k^{n-k}}.$$

Suppose now that $n - k = 1$, i.e., that the number of dimensional parameters of the problem exceeds the number of independent dimensions by one. There then exists just one similarity parameter ξ, and we have

$$g = g_1 \widetilde{G}(\xi).$$

In this case we call the problem self-similar. Introducing the flow variables in this form reduces the hydrodynamic equations from a system of partial differential equations to a set of ordinary differential equations.

Let us now consider our problem. We choose a system of units of measurement with $k = 3$ independent units of mass $[M]$, length $[L]$, and time $[\theta]$. The hydrodynamic equations depend on three-dimensional governing parameters, namely m with $[m] = ML^{-2}$, t with $[t] = \theta$, and a with $[a] = M^{\mu+1} L^{-2\lambda - 3\mu - 1} \theta^{2\lambda - 1}$. Additional parameters come from the boundary conditions.

As a boundary condition at the vacuum–material interface, we specify the specific internal energy in the first Lagrangian cell of the body. This boundary condition is entirely equivalent to a boundary condition for the temperature through Eqs. (4) and (5); the internal energy is preferred only because we formulate the problem in mechanical units. We give the boundary condition for a modified internal energy $W = (\gamma - 1)e$, allowing for a time-dependence, in the form

$$\begin{aligned} (pv)|_{m=0} &= (\gamma - 1)e_v = W_v = W_v^* t^{\tau}; \\ [W_v^*] &= L^2 \theta^{-(2+\tau)}. \end{aligned} \tag{8}$$

The subscript v denotes the vacuum–material interface. This boundary condition adds another dimensional W_v^* (note that its dimension depends on τ). We then have $n = 4$ dimensional parameters $a_1 = t$, $a_2 = a$, $a_3 = W_v^*$, and $a_4 = m$. Because $n - k = 1$ we can form only a single dimensionless variable ξ_1 given by

$$\xi_1 = m\, a^{\alpha_1} t^{\tau_1} W_v^{*\omega_1}. \tag{9}$$

If the problem contained no other dimensional parameter it would obviously be self-similar.

Unfortunately the problem of heating of a solid body contains at least another dimensional parameter, namely the initial density ρ_s of the body. This parameter enters the problem through the boundary conditions at the interface with the solid. As long as it is important, the problem is not self-similar.

D. Asymptotic self-similar solution

Of central importance for the present investigation is the observation that in the ablative heat wave regime, the initial density loses its importance as a governing parameter of the problem as time goes to infinity. In this limit an asymptotic self-similar solution exists for the general case of a time-varying boundary temperature.

Let us first present a physical argument in support of this observation. The heat supplied at the vacuum boundary has to diffuse through an increasing mass of hot, expanding plasma in order to heat new layers of cold material in the interior part of the body. Thus, if the boundary temperature is kept constant, the rate of plasma production will slow down with time. On the other hand, the characteristic expansion velocity of the plasma, namely the sound velocity corresponding to the boundary temperature, remains constant. Hence, by virtue of mass conservation, the characteristic plasma density must decrease with time and will therefore become much less than ρ_s. One might therefore expect that, as time goes to infinity, the hydrodynamic flow of the conduction heated material becomes independent of the numerical value of ρ_s.

In order to elaborate this argument further, let us form the characteristic plasma density ρ_1 by dimensional analysis

$$\rho_1(t,a,W_v^*) = \left(\frac{t^{1-\tau(\lambda-1)}}{aW_v^{*\lambda-1}}\right)^{-1/(\mu+1)}. \tag{10}$$

As we shall see later (see Sec. IV B) this density corresponds, within a factor of order unity, to the density at the sonic point. We are interested in the heating (rather than cooling) of the body, i.e., in the case $\tau \geqslant 0$. If the time exponent of ρ_1 is inspected, it is found that $\rho_1 \to 0$ for $t \to \infty$ in the range $0 \leqslant \tau < \tau_M \equiv 1/(\lambda - 1)$. The characteristic plasma density becomes with time much less than the solid density not only for a constant boundary temperature, but for a variable one in this range of τ.

In order to assess the role of the parameter ρ_s in the limit $t \to \infty$, we consider the plasma–solid interface. As shown in Fig. 1(c), the conduction heated plasma is connected with the undisturbed solid through a shock wave. The shock wave takes up the whole recoil momentum of the expanding plasma but only a fraction of the order $(\rho_1/\rho_s)^{1/2}$ of the convective energy flux of the plasma. If $\rho_1 \to 0$ for $t \to \infty$, the ablation of material occurs eventually like from a rigid wall of infinite density which takes up momentum, but no energy; the heat flux supplied at the vacuum boundary is then entirely transformed into convective and internal energy of the plasma.

In this limit the hydrodynamic flow of the conduction heated material does no longer depend on the properties of the shock wave region, and may be calculated by simplified boundary conditions at the plasma–solid interface corresponding to infinite density, zero temperature, and zero velocity (the pressure is of course finite, but can be determined only through the solution of the problem). Most important, these boundary conditions do not contain any dimensional parameter, in particular no longer the parameter ρ_s. The asymptotic ablative heat wave therefore depends indeed only on the four parameters t, a, W_v^*, and m and is self-similar.

The self-similar solution obtained in this way will approach the exact solution of the problem everywhere in the heated material except for a thin layer near the ablation front where the density in the similarity solution exceeds solid density. However, as long as one is not specially interested in the details of this transition layer, it can be ignored because its fractional mass and energy content tend to zero with time.

E. The equivalence of different boundary conditions

In the preceding section we considered the heating of a solid body for the case of a given time-dependent boundary temperature. Indeed we will solve the problem in this form in the main part of the paper.

However, sometimes the problem may be posed in a different manner. For example, one might be interested in a self-similar solution where the heat flux into the plasma is given as a boundary condition. Other examples of possible interest are solutions where the pressure exerted onto the solid is specified or where the characteristic plasma density remains at a fixed value independent of time. Each of these solutions would be characterized by a dimensional parameter, called Q in the following, which is different in each case. For the examples mentioned, Q would be a heat flux S_0, a pressure p_0, and a density ρ_0, respectively. Here the subscript 0 is used to distinguish these parameters from the corresponding variables.

We shall show now that characterization of the problem by such a parameter Q is equivalent to a time-dependent boundary temperature with a certain time exponent τ. Physically this means that a condition imposed onto the solution may be met if the boundary temperature is suitably "programmed."

The equivalent τ for a given Q can be found as follows. Formally the parameter Q (Q plays the role of an additional parameter a_5) introduces a second dimensionless similarity parameter

$$\xi_2 = Q\, a^{\alpha_2}\, t^{\tau_2}\, W_v^{*\omega_2} \tag{11}$$

into the problem. Here τ is now determined in such a manner that the problem remains self-similar. This is possible if, through the proper choice of τ, ξ_2 is made a time-independent constant rather than a variable. In this case, because ξ_2 does not depend on m and t, the partial derivatives $\partial\xi_2/\partial m$ and $\partial\xi_2/\partial t$ disappear and hence the dimensionless hydrodynamic equations remain unchanged (compare Sec. III A) and are, in particular, a function of the single variable ξ_1 (the constant ξ_2 whose value depends on the numerical value of Q, appears as a parameter in the functions $\widetilde{G}$ only). With the dimensions of Q taken as $[Q] = M^{q_1} L^{q_2} \theta^{q_3}$, with the dimensions of W_v^* given by Eq. (8), and with $\tau_2 = 0$ this condition, applied to Eq. (11), yields readily the desired equivalence between τ and Q through dimensional analysis:

$$\tau = \frac{-2q_1(3\mu+2) - 2q_2(\mu+1) - 2q_3(\mu+1)}{q_1(2\lambda+3\mu+1) + q_2(\mu+1)}. \tag{12}$$

We shall solve in this paper the ablative heat wave problem for four discrete values of τ, equivalent to the dimensional parameters W_0 (resp. T), S_0, p_0, and ρ_0 (see Sec. IV). [The corresponding values of τ from Eq. (12) may be found in Tables II and III.]

The equivalence of τ and Q is of interest also with respect to previous work on the subject (see Sec. I F). A problem that has one governing parameter too many can obviously be made self-similar by imposing a certain time-dependence on another parameter of the problem (we chose the boundary temperature). Thus the benefit of self-similarity is achieved by sacrificing the free choice of the time dependence of another governing parameter of the problem. In this case, only a special self-similar solution is obtained which, in general, will not correspond to the boundary conditions of interest.

F. Previous work

As outlined above, self-similarity is achieved if one of the five (including ρ_s) governing parameters of the problem is discarded. Various approaches have been taken in this respect.

Perhaps the most widely known self-similar solution is the thermal wave in a medium of constant density. Because the density in this case does not depend on m and t, it is readily seen from the energy equation that the problem only depends on a single parameter $a\rho_s^{1-\mu}$ and not on a and ρ_s separately. Hence the number of dimensional parameters is reduced by one and the problem is self-similar. This solution is described in detail in Refs. 1 and 5. It is not applicable to the ablative heat wave where expansion is important and the density is not constant.

The possibility that a special self-similar solution of the ablative heat wave problem may be obtained if the solid density is kept as a governing parameter was apparently pointed out first by Marshak.[5] He noticed that, for the case of radiation heat conduction in a fully ionized plasma, the problem is self-similar if the boundary temperature varies as $T_v \sim t^{1/5}$. A similar approach was taken by Anisimov[6] in the treatment of laser heating of a solid target where electronic heat conduction is dominant. Here the temporal variation of the boundary temperature required for self-similarity is $T_v \sim t^{2/3}$. An extensive treatment of the laser heating temperature problem based on this type of self-similar solutions was recently given by Barrero and Sanmartín.[7-9]

Both time exponents are readily obtained from our considerations in Sec. I E. With $Q=\rho_s$, i.e., $[Q]=[\rho_s]=ML^{-3}$ and $q_1=1, q_2=-3, q_3=0$, one obtains $\tau=1/(\lambda-1)$ from Eq. (12). In this case $\tau=\tau_M$, i.e., the case $Q=\rho_s$ is the limiting case discussed in Sec. I D where the characteristic plasma density becomes independent of time. For radiation heat conduction in a fully ionized plasma with $\delta=1$ and $\nu=6$, one obtains $\lambda=(\nu+1-\delta)/\delta=6$ and $\tau=1/5$ (Marshak's case). For electronic heat conduction in a fully ionized plasma ($\delta=1$, $\nu=5/2$) one gets $\tau=2/3$ (Anisimov's case).

Another method of obtaining a self-similar solution of our set of fundamental equations is to eliminate time; in this way a stationary solution is obtained. In fact, approximate stationary solutions taking expansion into account have been given for the case of laser heating[10] and heating by thermal radiation.[11] However, these solutions do not apply to a truly planar case, which is necessarily nonstationary and miss important aspects of the problem connected with the time dependence of the physical quantities.

A common feature in Refs. 5–9 is that the solid density ρ_s is kept as a governing parameter of the problem. This leads to the restrictions imposed onto the self-similar solutions mentioned before, like the absence of expansion or a prescribed temporal variation of the boundary condition. However, as we have discussed in Sec. I D, ρ_s eliminates itself with time and is not really an essential parameter of the ablative heat wave problem. The previous treatments appear, therefore, unnecessarily restrictive.

To our knowledge, Afanas'ev *et al.*[12] used for the first time the assumption of an infinitely dense body to simplify the somewhat related problem of laser heating of a solid target in the optically thin regime. This idea is exploited here in a systematic manner for the problem of the heating of a body by nonlinear conduction. The observation that the ablative heat wave is asymptotically self-similar offers us the possibility to obtain self-similar solutions for a whole range $(0<\tau<\tau_M)$ of the time exponent τ (actually the scaling laws for the various hydrodynamic quantities can be obtained immediately from dimensional analysis, even if the problem is not solved exactly). In addition, the problem is conceptually simplified to a great extent. For further discussions we refer to Sec. VI.

II. DIMENSIONAL ANALYSIS

We now proceed with standard dimensional analysis as briefly described in Sec. I C. The dimensions of the quantities which appear in the hydrodynamic equations and in the boundary condition are $[t]=\theta$, $[a]=M^{\mu+1}L^{-(2\lambda+3\mu+1)}\times\theta^{2\lambda-1}$, $[W_v^*]=L^2\theta^{-(2+\tau)}$, $[m]=ML^{-2}$, $[\gamma]=1$, $[v]=M^{-1}L^3$, $[p]=ML^{-1}\theta^{-2}$, $[u]=L\theta^{-1}$.

With $a_1=t, a_2=a, a_3=W_v^*, a_4=m$, we form a single dimensionless similarity parameter

$$\xi=mt^{\tau_\xi}a^{\alpha_\xi}W_v^{*\omega_\xi}. \tag{13}$$

We locate the plasma–vacuum interface at $\xi=0$, and the plasma–solid interface at $\xi=\xi_f$. The mass m_f between these boundaries is given by

$$m_f=\xi_f/(t^{\tau_\xi}a^{\alpha_\xi}W_v^{*\omega_\xi}),$$

whence the expression for ξ may be written as

$$\xi=\xi_f m/m_f. \tag{14}$$

This means that ξ and m are proportional to each other with a time-dependent proportionality factor. The variables u, v, p are written in the form

$$\begin{aligned} u(m,t)&=t^{\tau_u}a^{\alpha_u}W_v^{*\omega_u}\widetilde{U}(\xi)=u_1(t)\widetilde{U}(\xi),\\ v(m,t)&=t^{\tau_v}a^{\alpha_v}W_v^{*\omega_v}\widetilde{V}(\xi)=v_1(t)\widetilde{V}(\xi),\\ p(m,t)&=t^{\tau_p}a^{\alpha_p}W_v^{*\omega_p}\widetilde{P}(\xi)=p_1(t)\widetilde{P}(\xi). \end{aligned} \tag{15}$$

For use in the subsequent parts of this paper, we give explicitly the expressions for the temperature T, the heat flux S, and the total energy $E_{\rm tot}$ written in the same manner:

$$T(m,t)=T_1(t)\widetilde{T}(\xi);\quad \begin{cases} T_1(t)=(p_1v_1/K)^{1/\delta},\\ \widetilde{T}(\xi)=(\widetilde{P}\widetilde{V})^{1/\delta} \end{cases},$$

$$S(m,t)=S_1(t)\widetilde{S}(\xi);\quad \begin{cases} S_1(t)=a(p_1v_1)^{\lambda+1}v_1^{\mu-1}\xi_f/m_f,\\ \widetilde{S}(\xi)=(\widetilde{P}\widetilde{V})^{\lambda}\widetilde{V}^{\mu-1}\dfrac{\partial(\widetilde{P}\widetilde{V})}{\partial\xi} \end{cases},$$

$$E_{\mathrm{tot}}(t)=E_1(t)\widetilde{E}(\xi_f);\quad \begin{cases} E_1(t)=u_1^2 m_f/\xi_f, \\ \widetilde{E}(\xi_f)=\int_0^{\xi_f}\left(\dfrac{\widetilde{P}\widetilde{V}}{\gamma-1}+\dfrac{\widetilde{U}^2}{2}\right)d\xi. \end{cases}$$

The expressions for $E_{\mathrm{tot}}(t)$ are derived in Appendix A. There it is also shown that the ratio of the internal and kinetic energies is a time-independent constant in this problem.

We note that the value of the quantities at the boundary of the plasma with the vacuum and solid (i.e., at the front of the wave) are denoted by the subscripts v and f, respectively. For example, the temperature T_v at the vacuum boundary is given by

$$T_v \equiv T_1(t)\widetilde{T}(\xi=0)=T_1(t).$$

Here we have used the normalization

$$\widetilde{T}(\xi=0)=\widetilde{T}_v=1,$$

which is applied throughout this paper. The pressure p_f at the interface with the solid is

$$p_f=p_1(t)\widetilde{P}(\xi=\xi_f)=p_1(t)\widetilde{P}_f.$$

The exponents in Eqs. (13) and (14) are now determined by dimensional analysis. One finds for the similarity parameter

$$\xi=m/(a\, W_v^{*\lambda+(\mu-1)/2}\, t^{\mu+\tau[\lambda+(\mu-1)/2]})^{1/(\mu+1)},$$

for the dimensional part of the variables u, v, and p,

$$u_1(t)=W_v^{*1/2}\, t^{\tau/2},$$
$$v_1(t)=(a^{-1}W_v^{*1-\lambda}\, t^{1-\tau(\lambda-1)})^{1/(\mu+1)},$$
$$p_1(t)=(a\, W_v^{*\lambda+\mu}\, t^{\tau(\lambda+\mu)-1})^{1/(\mu+1)},$$

and for the dimensional part of the variables T, S, and E_{tot},

$$T_1(t)=T_v=(W_v^{*}\, t^{\tau}K^{-1})^{1/\delta},$$
$$S_1(t)=(aW_v^{*\lambda+(3\mu+1)/2}\, t^{\tau[\lambda+(3\mu+1)/2]-1})^{1/(\mu+1)}, \qquad (16)$$
$$E_1(t)=(aW_v^{*\lambda+(3\mu+1)/2}\, t^{\tau[\lambda+(3\mu+1)/2]})^{1/(\mu+1)}.$$

The exponents τ_ξ, τ_v, τ_p, and τ_u are given by

$$\tau_\xi=[-\mu-\tau(\lambda+\mu/2-1/2)]/(\mu+1),$$
$$\tau_v=[1-\tau(\lambda-1)]/(\mu+1),$$
$$\tau_p=[\tau(\lambda+\mu)-1]/(\mu+1),$$
$$\tau_u=\tau/2.$$

III. INTEGRATION OF THE HYDRODYNAMIC EQUATIONS

A. The hydrodynamic equations in dimensionless form

If the dependent variables are written in the form $g(m,t)=g_1(t)\widetilde{G}(\xi)$, and since

$$\frac{\partial g}{\partial m}=g_1\frac{d\widetilde{G}}{d\xi}\frac{\partial\xi}{\partial m},\quad \frac{\partial g}{\partial t}=\frac{dg_1}{dt}\widetilde{G}+g_1\frac{d\widetilde{G}}{d\xi}\frac{\partial\xi}{\partial t},$$

the hydrodynamic equations reduce from a system of partial differential equations to a system of ordinary differential equations:

$$-\tau_v\widetilde{V}-\tau_\xi\xi\frac{d\widetilde{V}}{d\xi}+\frac{d\widetilde{U}}{d\xi}=0, \qquad (17)$$

$$\tau_u\widetilde{U}+\tau_\xi\xi\frac{d\widetilde{U}}{d\xi}+\frac{d\widetilde{P}}{d\xi}=0, \qquad (18)$$

$$\left(\frac{\gamma\tau_v+\tau_p}{\gamma-1}\right)\widetilde{P}\widetilde{V}+\frac{\tau_\xi}{\gamma-1}\xi\frac{d(\widetilde{P}\widetilde{V})}{d\xi}+\tau_\xi\xi\widetilde{P}\frac{d\widetilde{V}}{d\xi} \qquad (19)$$
$$=\frac{d}{d\xi}\left((\widetilde{P}\widetilde{V})^{\lambda}\widetilde{V}^{\mu-1}\frac{d(\widetilde{P}\widetilde{V})}{d\xi}\right).$$

We have as boundary conditions at the plasma–vacuum boundary, $\xi=0$, $\widetilde{V}=+\infty$, $\widetilde{P}=0$, and $\widetilde{P}\widetilde{V}=1$, at the plasma–solid boundary $\xi=\xi_f$, $\widetilde{U}=0$, $\widetilde{V}=0$, and $\widetilde{P}=\widetilde{P}_f$. Note that ξ_f and $\widetilde{P}_f$ are *a priori* unknown.

Mathematically, Eqs. (17)–(19) are equivalent to a first-order system of five equations if $\widetilde{U}$, $\widetilde{V}$, $\widetilde{P}$, $d\widetilde{P}/d\xi$, and $d\widetilde{V}/d\xi$ are considered as independent variables (for numerical integration the equations were actually rewritten in this form). A complete set of boundary conditions, therefore, consists of initial values for all of these variables at a common ξ. Obviously, the physical boundary conditions noted above are incomplete and do not allow integration to be started, either at the plasma–vacuum or at the plasma–solid boundary. This difficulty can be overcome with the help of approximate solutions.

B. Approximate solutions

1. Approximate solution at the plasma–solid boundary

We try to find an analytical approximation for $\xi\to\xi_f$ of the form

$$\lim_{\xi\to\xi_f}\widetilde{P}(\xi)=\widetilde{P}_f;\quad \lim_{\xi\to\xi_f}\widetilde{V}(\xi)=\widetilde{V}_0(\xi_f-\xi)^{\beta}.$$

Making use of the boundary conditions, we find to a first-order approximation in $(\xi_f-\xi)$

$$\widetilde{U}=-\tau_\xi\xi_f\widetilde{V}_0(\xi_f-\xi)^{1/(\lambda+\mu-1)}, \qquad (20)$$
$$\widetilde{V}=\widetilde{V}_0(\xi_f-\xi)^{1/(\lambda+\mu-1)}, \qquad (21)$$
$$\widetilde{P}=\widetilde{P}_f-\tau_\xi^2\,\xi_f^2\,\widetilde{V}_0(\xi_f-\xi)^{1/(\lambda+\mu-1)}, \qquad (22)$$

where

$$\widetilde{V}_0=[-\gamma(\gamma-1)^{-1}(\lambda+\mu-1)\tau_\xi\widetilde{P}_f^{-\lambda}\xi_f]^{1/(\lambda+\mu-1)}.$$

It is interesting to note that we get for the exponent of $(\xi_f-\xi)$ the same value as Marshak[5] in his dimensionless solution for the problem with constant pressure.

2. Approximate solution at the plasma–vacuum boundary

For the case of a constant boundary temperature $(\tau=0)$, an approximate solution can also be given near the plasma–vacuum boundary. We look for solutions of the form

$$\lim_{\xi\to 0}\widetilde{V}(\xi)=\widetilde{V}_v\xi^{\beta_1};\quad \beta_1<0,$$

with

$$\lim_{\xi\to 0}\widetilde{P}(\xi)\widetilde{V}(\xi)=\widetilde{W}_v,$$

and

$$\lim_{\xi\to 0}(\widetilde{P}\widetilde{V})^{\lambda}\widetilde{V}^{\mu-1}\frac{d(\widetilde{P}\widetilde{V})}{d\xi}=\widetilde{S}_v.$$

Matching coefficients and exponents of ξ at both sides of Eqs. (17)–(19), we obtain in a third-order approximation,

$$\tilde{U} = \tilde{U}_0 + (\tau_v - \tau_\xi)\tilde{V}_1 \ln \xi + [\tau_v + (\mu - 1)\tau_\xi]\tilde{V}_2 \mu^{-1}\xi^{\mu} + (\tau_v + \mu\tau_\xi)(\mu + 1)^{-1}\tilde{V}_3 \xi^{\mu+1},$$

$$\tilde{V} = \tilde{V}_1 \xi^{-1} + \tilde{V}_2 \xi^{\mu-1} + \tilde{V}_3 \xi^{\mu},$$

$$\tilde{P} = \tau_\xi(\tau_\xi - \tau_v)\tilde{V}_1 \xi - \tau_\xi[\tau_v + (\mu - 1)\tau_\xi](\mu + 1)^{-1}\tilde{V}_2 \xi^{\mu+1} - \tau_\xi(\tau_v + \mu\tau_\xi)(\mu + 2)^{-1}\tilde{V}_3 \xi^{\mu+2},$$

where

$$\tilde{V}_1 = [\tilde{W}_v \tau_\xi^{-1}(\tau_\xi - \tau_v)^{-1}]^{1/2},$$

$$\tilde{V}_2 = \tilde{S}_v(\mu + 1)\tilde{V}_1^{-(2\lambda+\mu)}\mu^{-2}\tau_\xi^{-(\lambda+1)} \times (\tau_\xi - \tau_v)^{-\lambda}(2\tau_\xi - \tau_v)^{-1},$$

$$\tilde{V}_3 = \frac{[(\gamma\tau_v + \tau_p)/(\gamma - 1) - \tau_\xi](\mu + 2)[\tau_\xi(\tau_\xi - \tau_v)]^{\mu/2}}{\tilde{W}_v^{\lambda+\mu/2-1}(\mu + 1)\tau_\xi[2\tau_\xi - (\mu + 3)\tau_v]}.$$

If $\tilde{W}_v$ and $\tilde{S}_v$ are specified, the approximation solution is fully determined.

C. Numerical integration

For numerical integration, Eqs. (17)–(19) were written as a first-order system of five equations and numerically integrated on a desk-top computer by means of a standard Runge–Kutta routine.

Integration was started near the plasma–solid boundary with a set of initial values calculated from the approximate solutions (20)–(22). Because the scale factor ξ_f in Eqs. (20)–(22) is unknown, a preliminary coordinate ξ^* was chosen by setting $\xi_f^* = 1$. The second open parameter in Eqs. (20)–(22), the dimensionless pressure at the plasma–solid boundary $\tilde{p}_f$ [all quantities in the ξ^* frame are denoted by small characters, e.g., $\tilde{p}(\xi^*)$, etc.], was then determined numerically by successive approximations from the conditions that the solution has to fulfill the proper boundary conditions at the plasma–vacuum boundary. The procedure is illustrated in Fig. 2 for $\tilde{p}$, which has to fulfill the boundary condition $\tilde{p}_v = 0$. Once $\tilde{p}_f$ has been determined in this manner, numerical integration gives the solution in ξ^* space, including the values at the boundaries.

With the arbitrary choice of the ξ^* scale, one obtains, in general, a value $\tilde{w}_v \neq 1$ instead of the desired boundary condition $\tilde{W}_v = 1$ (i.e., $W = W_v$ at the vacuum boundary). This difficulty can be overcome by rescaling the solution with the help of a similarity transformation. Note that the boundary condition $\tilde{W}_v = \tilde{P}_v \tilde{V}_v = \tilde{T}_v^{\delta} = 1$ follows from our normalization $\tilde{T}_v = 1$.

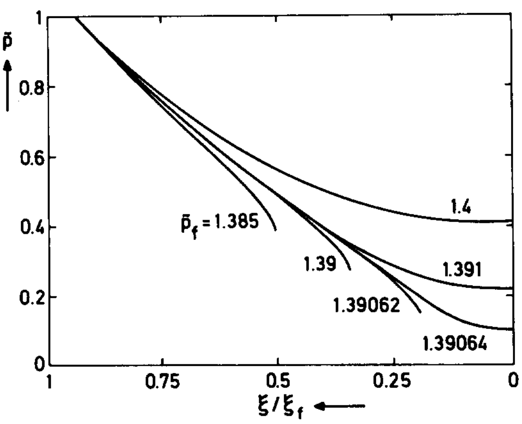

FIG. 2. Determination of the pressure at the solid–plasma boundary by numerical iteration.

As may be verified by insertion, the set of differential equations (17)–(19) admits a similarity transformation, i.e., it remains unchanged for a transformation $\xi_1 \to \xi$ of the form $\xi = c\xi_1$, provided the variables are also transformed according to

$$\tilde{V}(\xi) = \tilde{v}(\xi_1)c^{\beta_1}, \quad \tilde{U}(\xi) = \tilde{u}(\xi_1)c^{\beta_1+1}, \quad \tilde{P}(\xi) = \tilde{p}(\xi_1)c^{\beta_1+2}, \tag{23}$$

where c is an arbitrary constant and β_1 is given by

$$\beta_1 = 2(1 - \lambda)(2\lambda + \mu - 1)^{-1}.$$

To rescale the numerical results from the ξ^* to the ξ frame, the constant c is determined from the condition

$$\tilde{p}_v \tilde{v}_v c^{2(\beta_1+1)} = \tilde{P}_v \tilde{V}_v = 1,$$

i.e., from the numerical value of the product $\tilde{p}_v \tilde{v}_v$ determined by the integration in ξ^* space. The results of the numerical integration will be presented already in terms of the final variables $\tilde{U}(\xi)$, $\tilde{V}(\xi)$, and $\tilde{P}(\xi)$.

We note that the dimensionless hydrodynamic equations admit an analytical solution; it is given in Appendix B. It was used to control the accuracy of the numerical integration.

D. Transformation of the solutions into real space

Once the problem has been solved in dimensionless variables, the results have to be transformed back into real space. We take the origin of the spatial x coordinate at the plasma–solid boundary, with the positive x direction towards the vacuum. With the definition of the Langrangian mass coordinate $dm = v^{-1}\,dx$ and Eq. (14), one has

$$dx = -(m_f/\xi_f)v_1(t)\tilde{V}(\xi)d\xi.$$

Insertion of the expressions for m_f and ξ_f and integration yield

$$x = -\frac{m_f}{\xi_f}v_1(t)\int_{\xi_f}^{\xi}\tilde{V}(\xi')d\xi' = -\frac{W_v^{1/2}t}{\xi_f}\int_{\xi_f}^{\xi}\tilde{V}(\xi')d\xi'. \tag{24}$$

This relation may be written by introducing a variable

$$y \equiv xW_v^{-1/2}t^{-1}\xi_f = -\int_{\xi_f}^{\xi}\tilde{V}(\xi')d\xi'.$$

We call y the similarity coordinate in real space; it is related to ξ through an integral over the solution $\tilde{V}(\xi)$.

IV. RESULTS

A. Dimensional dependences

As already outlined in the introduction, we apply in this investigation a boundary condition of the form $W_v = (\gamma - 1)e_v = W_v^* t^{\tau}$. The exponent τ is related through Eq. (12) to a quantity Q which is a temporally constant governing parameter of the problem. In Table I we listed the dimensional part of the hydrodynamic variables for several choices of τ and hence Q. Here τ was chosen such that one obtains for Q either a temperature T_0 (related to W by $W_0 = KT_0^\delta$), heat flux S_0, pressure p_0, or density ρ_0.

The numerical examples calculated explicitly are for a wall of solid gold. For this case (see Appendix C) we chose $\lambda = 3, \mu = 2$. Table II gives the dimensional dependence for this case. It is seen, for example, that a constant temperature can be maintained by a decreasing flux $S_1 \sim t^{-1/3}$, a constant pressure requires only a slightly increasing flux $S_1 \sim t^{1/10}$, whereas a constant plasma density can only be maintained by a flux increasing according to $S_1 \sim t^{3/4}$.

We note that the quantities with the subscript 0 should be understood as characteristic parameters of the problem and the relations given in Tables I and II as scaling relations. The expressions still lack a numerical factor which depends on how the parameter Q enters through the boundary conditions and which can be determined only by integration of the hydrodynamic equations. In order to obtain, for example in the case $Q = S_0$, exact expressions as a function of the heat flux S_v at the vacuum boundary, one has to replace S_0 with $S_0 = S_v/\widetilde{S}_v$; this yields the numerical factor provided the boundary value $\widetilde{S}_v$ of the dimensionless function $\widetilde{S}$ has been determined. Some boundary values of the dimensionless functions will be given in Table III.

The scaling relations of Table I may be obtained directly from elementary diffusion theory (we are grateful to the referee of this paper that he emphasized this point). For example, for the case of constant boundary temperature and $\delta = 1$, one has $m^2 \propto T^\lambda v^{\mu - 1} t$. Eliminating v in the obvious way $v \propto c_u\ (dm/dt)$, where $c_u \propto T^{1/2}$ is the sound velocity, one finds $m \propto (T^{\lambda + (\mu - 1)/2} t^\mu)^{1/(\mu + 1)}$ as in Table I.

The possibility to derive the scaling laws in this way is of interest from a physical point of view because it shows clearly the role of the diffusion of radiation as the cause of the ablative flow. The derivation implies of course, the considerations, in particular with respect to ρ_s, given in the introduction and does by no means invalidate the dimensional analysis underlying Table I (which in fact is elementary as well). The formal similarity approach is indispensable in order to calculate the numerical factors in front of the scaling relations and, of course, the spatial flow profiles.

B. Numerical results

Let us discuss in some detail the numerical results obtained for the case of a constant boundary temperature ($\tau = 0$). The variation of the dimensionless hydrodynamic variables is shown in Fig. 3. The calculations (actually for all of the figures) were made with $\gamma = 3/2$, $\delta = 3/2$, $\lambda = 3$, and $\mu = 2$ (compare Appendix C).

The dimensionless temperature $\widetilde{T}$ and the dimensionless heat flux $\widetilde{S}$ steadily decrease from the vacuum boundary to the plasma–solid boundary, where they become zero. At the vacuum boundary one has $\widetilde{T}_v = 1$ (as specified by the boundary condition); the dimensionless heat flux there has the value $\widetilde{S}_v = 1.68$. The dimensionless pressure $\widetilde{P}$ is zero at the vacuum boundary and attains a finite value $\widetilde{P}_s = 0.758$ at the plasma–solid boundary (with the coordinate $\xi_f = 0.65$). The dimensionless flow velocity $\widetilde{U}$ is zero at the solid and diverges at the vacuum boundary. The dimensionless density $\widetilde{R}$ becomes infinite at the solid and zero at the vacuum boundary (with the corresponding behavior for $\widetilde{V}$).

The sonic point is located at a fixed value of ξ determined by the condition

TABLE I. General dimensional relations. Here λ and μ are related to j and μ', the exponents of the Rosseland mean free path $l_R \propto T^j v^{\mu'}$, by $\lambda = (j + 4 - \delta)/\delta$ and $\mu = \mu'$ [see Eq. (7) and Appendix C]. Here δ is the temperature exponent in the equation of state $pv = KT^\delta$ [Eq. (5)]. Note that W_0 and T_0 are related by $W_0 = KT_0^\delta$.

Q	τ	ξ	u_1	v_1	p_1
W_0	0	$m/(aW_0^{\lambda + (\mu - 1)/2} t^\mu)^{1/(\mu + 1)}$	$W_0^{1/2}$	$(a^{-1} W_0^{1 - \lambda} t)^{1/(\mu + 1)}$	$(aW_0^{\lambda + \mu} t^{-1})^{1/(\mu + 1)}$
S_0	$2/(2\lambda + 3\mu + 1)$	$m(a^2 S_0^{2\lambda + \mu - 1} t^{2\lambda + 3\mu - 1})^{1/(2\lambda + 3\mu + 1)}$	$(a^{-1} S_0^{\mu + 1} t)^{1/(2\lambda + 3\mu + 1)}$	$(a^{-3} S_0^{2 - 2\lambda} t^3)^{1/(2\lambda + 3\mu + 1)}$	$(aS_0^{2(\lambda + \mu)} t^{-1})^{1/(2\lambda + 3\mu + 1)}$
p_0	$1/(\lambda + \mu)$	$m/(ap_0^{2\lambda + \mu + 1} t^{2\lambda + 2\mu - 1})^{1/(2\lambda + 2\mu)}$	$(a^{-1} p_0^{\mu + 1} t)^{1/2(\lambda + \mu)}$	$(a^{-1} p_0^{1 - \lambda} t)^{1/(\lambda + \mu)}$	p_0
ρ_0	$1/(\lambda - 1)$	$m/(a^{-1} \rho_0^{2\lambda + \mu - 1} t^{2\lambda - 1})^{1/2(\lambda - 1)}$	$(a^{-1} \rho_0^{\mu + 1} t)^{1/2(\lambda - 1)}$	ρ_0^{-1}	$(a^{-1} \rho_0^{\lambda + \mu} t)^{1/(\lambda - 1)}$

TABLE II. Dimensional relations for our standard case $\delta = 3/2$ and $(j,\mu') = (2,2)$, corresponding to $(\lambda,\mu) = (3,2)$ (multiply ionized gold, see Appendix C).

Q	τ	ξ	u_1	v_1	p_1	E_1	S_1
W_0	0	$m/(aW_0^{7/2} t^2)^{1/3}$	$W_0^{1/2}$	$(a^{-1} W_0^{-2} t)^{1/3}$	$(aW_0^5 t^{-1})^{1/3}$	$(aW_0^{13/2} t^2)^{1/3}$	$(aW_0^{13/2} t^{-1})^{1/3}$
S_0	2/13	$m/(a^2 S_0^7 t^{11})^{1/13}$	$(a^{-1} S_0^3 t)^{1/3}$	$(a^{-3} S_0^{-4} t^3)^{1/13}$	$(aS_0^{10} t^{-1})^{1/13}$	$S_0 t$	S_0
p_0	1/5	$m/(ap_0^7 t^9)^{1/10}$	$(a^{-1} p_0^3 t)^{1/10}$	$(a^{-1} p_0^{-5} t)^{1/10}$	p_0	$(a^{-1} p_0^{13} t^{11})^{1/10}$	$(a^{-1} p_0^{13} t)^{1/10}$
ρ_0	1/2	$m/(a^{-1} \rho_0^7 t^5)^{1/4}$	$(a^{-1} \rho_0^3 t)^{1/4}$	ρ_0^{-1}	$(a^{-1} \rho_0^5 t)^{1/2}$	$(a^{-3} \rho_0^{13} t^7)^{1/4}$	$(a^{-3} \rho_0^{13} t^3)^{1/4}$

TABLE III. Calculated values for four similarity solutions in each of which a different parameter Q is held constant in time [Q is either a temperature T_0 (note $W_0 = KT_0^\delta$) a heat flux S_0, a pressure p_0, or a density ρ_0]. Here τ is the time exponent of the boundary temperature required to achieve a temporally constant Q, γ the adiabatic index. The values are given for four different pairs of exponents (j, μ') of the Rosseland mean free path $l_R \sim T^j v^{\mu'}$. Here ξ_f is the front coordinate of the wave (plasma–solid interface), the dimensionless temperature is normalized such that $(\widetilde{P}\widetilde{V})_v = T_v^\delta = 1$. The boundary values for the dimensionless heat flux $\widetilde{S}_v$ (the index v denotes the plasma–vacuum boundary), the dimensionless pressure $\widetilde{P}_f$ at the wave front, and in addition, the coordinate of the sonic point $\xi_{M=1}$ and the dimensionless density $\widetilde{R}_{M=1}$ at the sonic point are given. Here $\delta = 3/2$ always.

Q	j	μ'	τ	γ	ξ_f	$(\widetilde{P}\widetilde{V})_v$	$\widetilde{S}_v$	$\widetilde{P}_f$	$\xi_{M=1}$	$\widetilde{R}_{M=1}$
	1	2			0.59		2.15	0.67	0.17	0.43
T_0	2	2	0	5/4	0.56	1	2.03	0.65	0.32	0.41
	1	1			0.53		1.55	0.44	0.19	0.31
	2	1			0.36		0.93	0.29	0.17	0.12
	1	2			0.51		2.47	0.71	0.31	0.44
S_0	2	2	2/13	5/4	0.48	1	2.39	0.70	0.34	0.42
	1	1			0.46		2.07	0.53	0.23	0.33
	2	1			0.30		1.16	0.37	0.17	0.11
	1	2			0.50		2.55	0.72	0.16	0.44
P_0	2	2	1/5	5/4	0.46	1	2.49	0.71	0.32	0.43
	1	1			0.45		2.21	0.55	0.18	0.33
	2	1			0.28		1.22	0.35	0.10	0.11
	1	2			0.41		3.02	0.78	0.14	0.45
ρ_0	2	2	1/2	5/4	0.37	1	2.96	0.78	0.26	0.45
	1	1			0.27		1.63	0.43	0.17	0.30
	2	1			0.22		1.56	0.42	0.09	0.10
T_0	2	2	0	3/2	0.65	1	1.68	0.76	0.47	0.48

$$M = \frac{u}{C_u} = \frac{W_v^{1/2}\widetilde{U}(\xi)}{W_v^{1/2}\widetilde{W}(\xi)^{1/2}} = \frac{\widetilde{U}(\xi)}{\widetilde{W}^{1/2}(\xi)} = 1.$$

Its numerical value is $\xi_{M=1} = 0.47$. The density at the sonic point is

$$\rho_{M=1} = \rho_1(t)\widetilde{R}(\xi_{M=1}) = 0.48\rho_1(t).$$

This relation illustrates the physical meaning of the characteristic plasma density $\rho_1(t)$. It is, within a factor of order unity, equal to the density at the sonic point.

The degree to which the approximate solutions approach the numerical ones may be judged from Fig. 4.

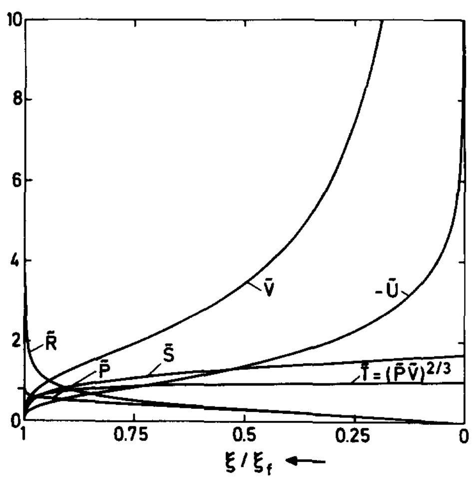

FIG. 3. Spatial variation of the dimensionless hydrodynamic variables in the conduction heated plasma. The solid–plasma interface is at $\xi/\xi_f = 1$, the plasma–vacuum interface at $\xi/\xi_f = 0$.

The curve y vs ξ, which affords transformation of the results into real space, is shown in Fig. 5. Figure 6(a) shows the curves of Fig. 3 in y space. From these graphs the values of the gasdynamic quantities can be evaluated for a given x and t.

In this representation it is also readily seen that the flow

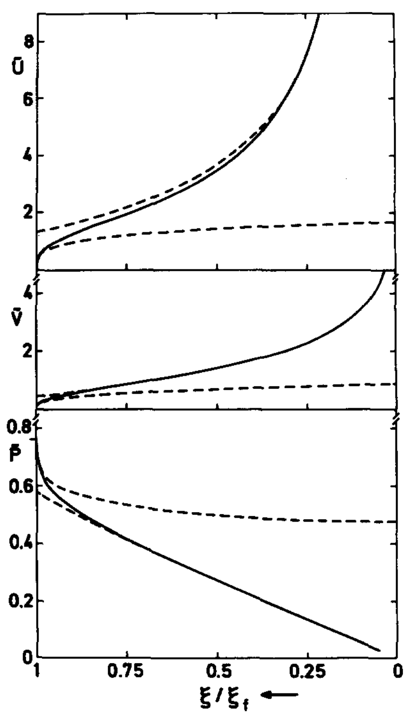

FIG. 4. Comparison of the analytical approximate solutions (dashed lines) with the numerical one (solid lines).

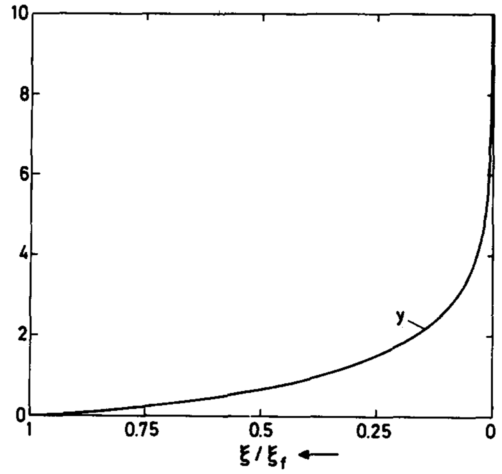

FIG. 5. The dimensionless variables y (the similarity variable in real space) vs ξ/ξ_f.

far from the solid approaches an isothermal rarefaction (linear increase of $\widetilde{U}$ with y), as one might expect for a constant boundary temperature.

Figure 6(b) shows, in addition, the dimensionless specific kinetic energy $\widetilde{E}_k$, the dimensionless specific internal energy $\widetilde{E}_i$, and, in addition to $\widetilde{U}$, the dimensionless sound velocity $\widetilde{C}_u = (\widetilde{P}\cdot\widetilde{V})^{1/2}$. The sonic point is located at $y = 0.26$.

The other case given in Table II, corresponding to the characteristic parameters S_0, P_0, ρ_0, were also integrated numerically. The shape of the calculated curves is, however, very similar to the case with a constant boundary temperature; they are therefore not reproduced here. Instead we give calculated values and the coordinates of the plasma–solid boundary and of the sonic point in Table III.

The calculations for Table III were performed for our standard case $(\lambda,\mu) = (3,2)$ and, because the exponents λ and μ depend on conditions, for three other pairs of exponents as well. We remember that λ and μ [compare Eq. (3)] are related to j and μ', the exponents of the Rosseland mean free path $l_R \sim T^j v^{\mu'}$, by $\lambda = (j + 4 - \delta)/\delta$ and $\mu = \mu'$ [see Eq. (7) and Appendix C]. Here δ is the temperature exponent in the equation of state $pv = KT^\delta$ [Eq. (5)], and $\delta = 3/2$ was chosen for all cases calculated. Besides the standard case $(\lambda,\mu) = (3,2)$, corresponding to $(j,\mu') = (2,2)$, the pairs $(j,\mu') = (2,1)$, $(1,2)$, and $(1,1)$ were chosen. At low temperature and high density, the case $(2,2)$ will tend to the case $(1,1)$, the latter corresponding to the maximum opacity possible (see Bernstein and Dyson Maximum Opacity Theorem[13]). The cases $(2,1)$ and $(1,2)$ are intermediate.

The case $Q = \rho_0$ corresponds to Marshak's case. In this case $\xi_2 = \rho_s/\rho_1(t) = \rho_s/\rho_0$ is a time-independent constant. The boundary condition of an infinitely dense body is appropriate in the limit $\rho_0 \ll \rho_s$ or $\xi_2 \gg 1$; this corresponds to the limiting case $\alpha \ll 1$ calculated by Sanmartín and Barrero[7–9] (their α is related to ξ_2 by $\xi_2 \sim \alpha^{-3/2}$). From a mathematical point of view, their calculations appear very similar to ours (neglecting the fact that they deal with electron heat conduction and take the critical density of laser radiation into account); the main difference lies in the fact that in the present work the restriction on the time-variation of the boundary temperature is removed.

Because Marshak's case has found some attention in the literature, it may be useful to complement Table II by Tables IV and V. Both tables are for a fully ionized plasma ($\delta = 1$), Table IV for radiation heat conduction ($\nu = 6, \mu = 2$), Table V for electron heat conduction ($\nu = 5/2, \mu = 0$). The choice of exponents is as in the papers by Marshak[5] and Anisimov,[6] respectively.

V. VALIDITY OF THE RESULTS

A. Validity of the diffusion approximation

For the diffusion approximation to be valid, the radiation mean free path l_R must be small compared with the temperature gradient length l_T, i.e., we must have locally,

$$[l_R(m,t)/l_T(m,t)] \ll 1. \tag{25}$$

From the definition of l_T we have

$$l_T^{-1} \equiv \frac{1}{T}\frac{\partial T}{\partial x} = \frac{1}{Tv}\frac{\partial T}{\partial m}$$

$$= \frac{1}{v_1(t)}\frac{\xi_f}{m_f}\frac{1}{\widetilde{T}\widetilde{V}}\frac{d\widetilde{T}}{d\xi} \equiv l_{1T}^{-1}(t)\widetilde{L}_T^{-1}(\xi), \tag{26}$$

where we have written l_T as a product of a dimensional and a dimensionless factor:

$$l_{1T}(t) = v_1(t)m_f/\xi_f = W_v^{1/2}t, \tag{27}$$

$$\widetilde{L}_T(\xi) = \frac{1}{\widetilde{T}(\xi)\widetilde{V}(\xi)}\frac{d\widetilde{T}}{d\xi}. \tag{28}$$

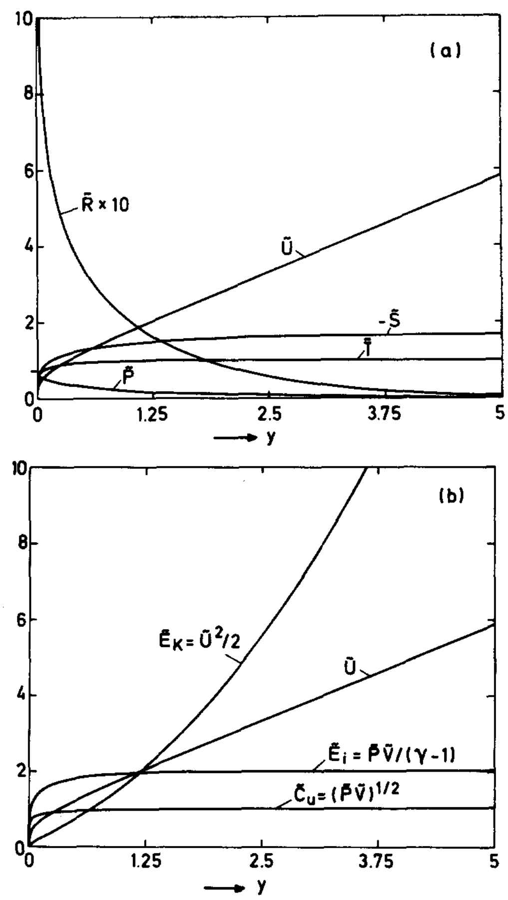

FIG. 6. Graphs of the numerical solutions versus y (i.e., in real space).

TABLE IV. Dimensional relations for $Q=\rho_0$ (Marshak's case) for radiation heat conduction in a fully ionized plasma ($\delta = 1$), and $(j,\mu') = (3,2)$ as assumed by Marshak.[5]

ξ	u_1	v_1	p_1	E_1	s_1	T_1
$m/(aW_0^{11/2}t^{16/5})^{1/3}$	$W_0^{1/2}t^{1/8}$	$(aW_0^5)^{-1/3}$	$(aW_0^8t^{3/5})^{1/3}$	$(aW_0^{19/2}t^{39/10})^{1/3}$	$(aW_0^{19/2}t^{19/10})^{1/3}$	$W_0K^{-1}t^{1/5}$

TABLE V. Dimensional relations for the case $Q=\rho_0$ for electronic heat conduction in a fully ionized plamsa ($\delta = 1$). For electronic heat conduction, $(\nu,\mu) = (5/2,0)$; ν and μ are the exponents of thermal conductivity $k \propto T^\nu v^\mu$. This case was treated by Anisimov.[6]

ξ	u_1	v_1	p_1	E_1	S_1	T_1
$m/(aW_0^2t^{4/3})$	$W_0^{1/2}t^{1/3}$	$a^{-1}W_0^{-3/2}$	$aW_0^{5/2}t^{2/3}$	$aW_0^3t^2$	aW_0^3t	$W_0K^{-1}t^{2/3}$

We write l_R, which we assume to be of the form $l_R = AT^jv^\mu$, in a similar manner:

$$l_R(m,t) = AT_1^jv_1^\mu\widetilde{T}^j(\xi)\widetilde{V}^\mu(\xi) \equiv l_{1R}(t)\widetilde{L}_R(\xi). \tag{29}$$

For l_R/l_T we then have

$$\frac{l_R}{l_T} = \frac{l_{1R}(t)}{l_{1T}(t)}\frac{\widetilde{L}_R(\xi)}{\widetilde{L}_T(\xi)}. \tag{30}$$

Here $\widetilde{L}_R/\widetilde{L}_T$ was calculated from the numerical solutions (see Fig. 7). In contrast to the dimensionless mass within a radiation mean free path $\widetilde{M}_R = \widetilde{R}\widetilde{L}_R$, which increases towards the vacuum, $\widetilde{L}_R/\widetilde{L}_T$ is everywhere of order unity (except in the immediate vicinity of the plasma–solid boundary). The condition (25) therefore reduces to the more global condition

$$l_R/l_T \approx l_{1R}(t)/l_{1T}(t) \ll 1,$$

which only depends on the boundary temperature. With the definitions of l_{1R} and l_{1T} it can be written as

$$T_v^{j-\delta/2-[(\lambda-1)/(\mu+1)]\mu'\delta}t^{\mu'/(\mu+1)-1}$$
$$\times A_lK^{-1/2-[(\lambda-1)/(\mu+1)]\mu'}a^{-\mu'/(\mu+1)} \ll 1.$$

For our example (gold) with $\delta = 3/2$, $j = 2$, $\lambda = 3$, $\mu' = \mu = 2$, it becomes

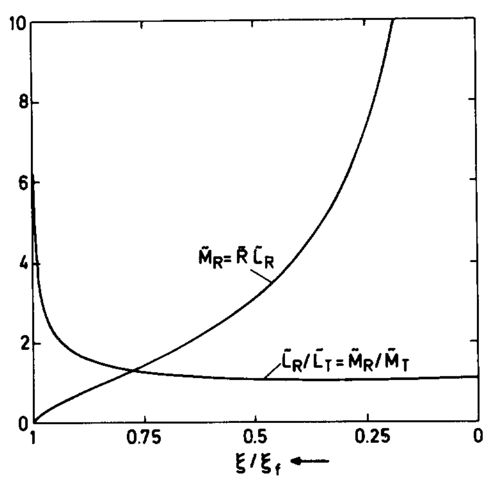

FIG. 7. The dimensionless mass per unit area within a radiation mean free path $\widetilde{M}_R$ and the ratio of the dimensionless radiation mean free path to the dimensionless temperature gradient length $\widetilde{L}_R/\widetilde{L}_T$ as a function of ξ/ξ_f.

$$T_v^{9/4}t \gg A_l^3a^{-2}K^{-11/2} = 1\times10^2 \text{ sec } {}^\circ\text{K}^{9/4}. \tag{31}$$

We note that as an alternate condition for the validity of the diffusion approximation one may take $S_1 \ll \sigma T^4$. With the definition of S_1 [see Eq. (16)] it becomes for gold,

$$T_v^{9/4}t \gg a\,K^{13/2}\sigma^{-3} = 4.4\times10^3 \text{ sec } {}^\circ\text{K}^{9/4}. \tag{32}$$

It is identical with (31) except for the numerical coefficient.

Condition (31) [resp. (32)] will be discussed together with the following condition (33) in Sec. VI.

B. The assumption of an infinitely dense body

Self-similarity was achieved in our approach by the condition $\rho_1(t) \ll \rho_s$. With the definition of ρ_1 [see Eq. (10)] it may be written as

$$(T_v^{(\lambda-1)\delta}t^{-1}K^{\lambda-1}a)^{1/(\mu+1)}\rho_s^{-1} \ll 1.$$

For our example it becomes

$$T_v^3t^{-1} \ll K^{-2}a^{-1}\rho_s^3 = 2.3\times10^{30}\text{sec}^{-1}\,{}^\circ\text{K}^3. \tag{33}$$

C. The assumption of negligible radiation pressure and energy

The radiation pressure is negligible when

$$f_R\sigma T^4(m,t)/c_0 \ll p(m,t), \tag{34}$$

where σ is the Stefan–Boltzmann constant, c_0 the light velocity, and $f_R = 4/3$.

The radiation energy is negligible when inequality (34) is fulfilled with $f_R = 4(\gamma - 1)$.

Taking into account that $p(m,t) = p_1(t)\,\widetilde{P}(\xi)$ and $T(m,t) = T_1(t)\,\widetilde{T}(\xi)$, (34) leads to

$$T_1 \ll \left[\left(\frac{aK^{\lambda+\mu}}{t}\right)^{1/(\mu+1)}\frac{c_0}{\sigma f_R}\frac{\widetilde{P}(\xi)}{\widetilde{T}^4(\xi)}\right]^{(\mu+1)/[4(\mu+1)-\delta(\lambda+\mu)]}, \tag{35}$$

or

$$T_1 \ll \left[\left(\frac{aK^5}{t}\right)^{1/3}\frac{c_0}{\sigma f_R}\frac{\widetilde{P}(\xi)}{\widetilde{T}^4(\xi)}\right]^{2/3}$$
$$= 3.21\times10^6t^{-2/9}\left(\frac{\widetilde{P}(\xi)}{f_R\widetilde{T}^4_{(\xi)}}\right)^{2/3}.$$

Now, with the approximations $\widetilde{P}(\xi)\cong(2/3)^{1/2}\xi$ (from Sec. III B for gold) and $\widetilde{T}(\xi)\cong 1$ for $\xi \ll 1$, we obtain from (35), taken with the equal sign, the coordinate ξ_R where radiation and material pressure become equal:

$$\xi_R = f_R \times 2.13\times 10^{-10} T_v^{3/2}(^\circ\mathrm{K}) t^{1/3}(\mathrm{sec}).$$

With $T_v = 10^6$ °K and $t = 10^{-10}$ sec, one finds $\xi_R \cong f_R \times 10^{-4}$.

For $\xi < \xi_R$ the radiation pressure and energy cannot be neglected. However, for the example given, the region $0<\xi<\xi_R$ contains only $\sim 10^{-4}$ of the ablated mass, hence the main body of the plasma is not affected. For not too long periods of time the assumption of negligible radiation pressure and energy is therefore well justified.

VI. DISCUSSION

A convenient way to discuss several features of interest and, in particular, the range of validity of the self-similar solutions calculated in this paper is afforded by Fig. 8. In this figure, the shaded range of validity is bounded by the three straight lines $l_{1R} = l_{1T}$, $\rho_1 = \rho_s$, and $p_1 = p_R$ (p_R denotes the radiation pressure). They correspond, respectively, to Eqs. (31), (33), and (35) [the last one with $\widetilde{P}(\xi)/\widetilde{T}^4(\xi) = 1$ and $f_R = 4/3$] taken with the equal sign. Note that in this double-logarithmic T-t diagram, a straight line corresponds to a time variation of the temperature in the form of a power law where the time exponent determines the slope of the line.

The line $\rho_1 = \rho_s$ separates the heat wave regime (located above) from the ablative heat wave regime. Its slope is given by $\tau = \tau_M$, i.e., by the exponent corresponding to Marshak's case. From the graphic representation it becomes quite clear that this case is an intermediate case between the two regimes: if the boundary temperature is varied by a power law $T_v \sim t^\tau$ with $\tau > \tau_M$ (i.e., with a slope larger than in the Marshak case), the flow will develop into the heat wave regime, for $\tau < \tau_M$ into the ablative heat wave regime. If $\tau = \tau_M$, the temperature follows in parallel the line $\rho_1 = \rho_s$; the flow is either of the heat wave or ablative heat wave type, depending on whether $\rho_1 \gtrless \rho_s$. If for the case $\tau < \tau_M$, the flow has developed into the ablative heat wave regime, it will finally reach the line $p_1 = p_R$, whence radiation pressure becomes important. However, this occurs after times longer than those of interest in laboratory experiments.

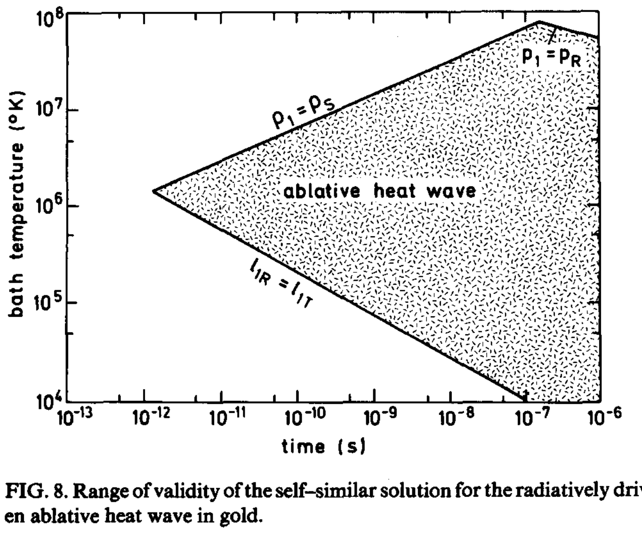

FIG. 8. Range of validity of the self-similar solution for the radiatively driven ablative heat wave in gold.

Below the line $l_{1R} = l_{1T}$, the radiation field and the heated material are not in local thermodynamic equilibrium, and the conduction approximation is not valid. In this regime, the radiation field is strongly anisotropic, dominated by photons traveling from the hot bath (now better called source) to the cold body (if the conduction approximation is valid the fluxes from the bath to the plasma and from the plasma to the bath nearly cancel each other). Simultaneously with the heating, expansion of the heated material takes place. As the material heats up and its characteristic temperature approaches the source temperature, gradually equilibrium between radiation and matter is established. One would expect that the transition to the optically thick ablative heat wave occurs at a time which in order of magnitude is given as a function of the source temperature by the line $l_{1R} = l_{1T}$ [the parallel line defined by the alternate condition Eq. (32) gives somewhat longer times]. This time increases rapidly as the source temperature decreases, a consequence of the fact that the plasma has time to expand and diminish its optical thickness if heating occurs by a cold and, hence, low-intensity source (keeping in mind that the source intensity varies as T^4). We note that this regime has some similarity with laser heating of a body in the optically thin regime.[12] A more detailed discussion of the processes occurring outside the shaded area will be given elsewhere.

An essential part of our results are the scaling relations for the ablative heat wave given in Sec. II and in Tables I and II. It may be useful to give explicitly the scaling laws and numerical values of some quantities for the heating of a gold wall in the case of a constant boundary temperature.

For the ablated mass, one has

$$m_f = \xi_f a^{1/3} K^{7/6} T_v^{7/4} t^{2/3}$$
$$= 8.2\times 10^{-8} T^{7/4}(^\circ\mathrm{K}) t^{2/3}(\mathrm{sec})\ \mathrm{g\ cm}^{-2}.$$

The characteristic plasma density ρ_1 is given by

$$\rho_1 = a^{1/3} K^{2/3} T_v t^{-1/3}$$
$$= 1.5\times 10^{-9} T_v(^\circ\mathrm{K}) t^{-1/3}(\mathrm{sec})\ \mathrm{g\ cm}^{-3}.$$

The pressure at the solid–plasma boundary is

$$p_f = \widetilde{P}_f a^{1/3} K^{5/3} T_v^{5/2} t^{-1/3}$$
$$= 8.2\times 10^{-6} T_v^{5/2}(^\circ\mathrm{K}) t^{-1/3}(\mathrm{sec})\ \mathrm{erg\ cm}^{-3}.$$

The heat flux at the plasma–vacuum boundary is

$$S_v = \widetilde{S}_v a^{1/3} K^{13/6} T_v^{13/4} t^{-1/3}$$
$$= 1.6\times 10^{-3} T_v^{13/4}(^\circ\mathrm{K}) t^{-1/3}(\mathrm{sec})\ \mathrm{erg\ sec}^{-1}\ \mathrm{cm}^{-2}.$$

For example, one obtains for $T_v = 5\times 10^6$ °K, $t = 10^{-9}$ sec, the following values: $m_f = 6.7\times 10^{-2}$ g cm^{-2} (corresponding to a layer thickness of 3.3×10^{-3} cm of solid gold), $\rho_1 = 7.3$ g cm^{-3}, $p_f = 4.6\times 10^{14}$ erg cm^{-3}, $S_v = 9.2\times 10^{21}$ erg sec^{-1} cm^{-2}. The transition from the heat wave to the ablative heat wave regime occurs when the line $\rho_1 = \rho_s$ is crossed. This occurs after $\sim 5\times 10^{-11}$ sec. It should be remembered, however, that in the derivation of the numerical value of a in Appendix C the (unknown) contribution of bound–bound transitions was not taken into account; thus the real values of a and of the quantities calculated above will be somewhat smaller.

Finally, we note that the energy flux required to achieve the conditions discussed in this paper, i.e., temperatures of $>10^6$ °K in an optically thick medium is of the order 10^{19} erg sec^{-1} cm^{-2} and above. Such fluxes are available from modern pulsed power generators, in particular, lasers. This now offers the possibility to study problems like the present one by experiments in the laboratory. A possible application is in inertial confinement fusion. The parameter range of inertial confinement fusion with temperatures of 10^6–10^7 °K and heating pulses of 10^{-9}–10^{-7} sec has considerable overlap with the area of validity of the present solutions.

ACKNOWLEDGMENTS

The authors would like to thank their colleagues J. Meyer-ter-Vehn and K. Unterseer for discussions, and S. Witkowski for valuable comments on the manuscript.

This work was supported in part by Euratom.

APPENDIX A: TOTAL ENERGY

The expression for the total thermal energy at time t is

$$E_T(t)=\int_{m=0}^{m_f(t)}\frac{p(m,t)v(m,t)}{\gamma-1}\,dm$$

$$=\frac{p_1(t)v_1(t)}{\gamma-1}\,\frac{m_f}{\xi_f}\int_{\xi=0}^{\xi_f}\widetilde{P}(\xi)\widetilde{V}(\xi)d\xi. \qquad \text{(A1)}$$

The expression for the kinetic energy is

$$E_K(t)=\int_{m=0}^{m_f(t)}\frac{u^2(m,t)}{2}\,dm$$

$$=\frac{u_1^2(t)}{2}\,\frac{m_f}{\xi_f}\int_{\xi=0}^{\xi_f}\widetilde{U}^2(\xi)d\xi. \qquad \text{(A2)}$$

It is easy to see that

$$p_1(t)v_1(t)=W_v^{*\omega_p+\omega_v}a^{\alpha_p+\alpha_v}t^{\tau_p+\tau_v},$$

$$u_1^2(t)=W_v^{*2\omega_u}a^{2\alpha_u}t^{2\tau_u}.$$

The dimensions of the product $p_1(t)v_1(t)$ and $u_1^2(t)$ are equal:

$$[p_1(t)v_1(t)]=[u_1^2(t)]=[\text{energy/mass}]=L^2\theta^{-2}.$$

Because only one combination of W_v^*, a, and t with these dimensions exists, it follows that

$$p_1(t)v_1(t)=u_1^2(t).$$

The relation between the kinetic and thermal energies at time t is

$$\frac{E_T(t)}{E_K(t)}=\frac{p_1(t)v_1(t)}{u_1^2(t)}\,\frac{2}{\gamma-1}\,\frac{\int_0^{\xi_f}\widetilde{P}(\xi)\widetilde{V}(\xi)d\xi}{\int_0^{\xi_f}\widetilde{U}^2(\xi)d\xi}=\frac{2}{\gamma-1}\,\eta(\xi_f),$$

where $\eta(\xi_f)$ is a numerical constant independent of time. We can define $E_1(t)$ as

$$E_1(t)=u_1^2(t)m_f/\xi_f=p_1(t)v_1(t)m_f/\xi_f, \qquad \text{(A3)}$$

and the total energy at time t takes the form

$$E_{\text{tot}}(t)=E_1(t)\left[\int_0^{\xi_f}\left(\frac{\widetilde{P}(\xi)\widetilde{V}(\xi)}{\gamma-1}+\frac{\widetilde{U}^2(\xi)}{2}\right)d\xi\right]$$

$$=E_1(t)\widetilde{E}(\xi_f), \qquad \text{(A4)}$$

where $\widetilde{E}(\xi_f)$ is a numerical constant.

APPENDIX B: ANALYTICAL SOLUTION OF THE DIMENSIONLESS HYDRODYNAMIC EQUATIONS

We find that

$$\widetilde{U}(\xi)=\widetilde{U}_0^*-\widetilde{U}_1^*\xi^{\beta+1},$$

$$\widetilde{V}(\xi)=\widetilde{V}_0^*\xi^{\beta},$$

$$\widetilde{P}(\xi)=\widetilde{P}_0^*\xi^{\beta+2},$$

are solutions of Eqs. (17)–(19) where

$$\beta=2(1-\lambda)/(2\lambda+\mu-1)$$

and

$$\widetilde{V}_0^*=\left(\frac{\gamma\tau_v+\tau_p+(3\beta+2)\tau_\xi}{2b^\lambda(\gamma-1)(\beta+1)[2(\lambda+1)(\beta+1)+\mu\beta-\beta-1]}\right)^{1/(2\lambda+\mu-1)},$$

$$\widetilde{P}_0^*=\widetilde{V}_0^*b,$$

$$b=-(\tau_v+\beta\tau_\xi)[(\beta+1)\tau_\xi+\tau_u](\beta+1)^{-1}(\beta+2)^{-1},$$

$$\widetilde{U}_1^*=-\widetilde{V}_0^*(\tau_v+\beta\tau_\xi)(\beta+1)^{-1}.$$

If $\tau_u=0$, the constant $\widetilde{V}_0^*$ may be chosen arbitrarily. If $\tau_u\neq 0$, the only way of matching both sides of Eq. (19) is by $\widetilde{U}_0^*=0$.

For this solution the integration (24) gives

$$x=u_1(t)t\int_{\xi'=0}^{\xi}\widetilde{V}(\xi')d\xi'=\widetilde{V}_0^*u_1(t)t\xi^{\beta+1}\beta^{-1},$$

and then

$$\xi=[x\beta\widetilde{V}_0^{*-1}u_1^{-1}(t)t^{-1}]^{1/(\beta+1)}=(y\beta\widetilde{V}_0^{*-1})^{1/(\beta+1)}.$$

APPENDIX C: EVALUATION OF THE CONSTANTS FOR RADIATION HEAT CONDUCTION IN A MULTIPLY IONIZED MATERIAL

The numerical results obtained in this paper were calculated with certain values for the constants and exponents appearing in the equations given in Sec. I B. These values were chosen in an attempt to model approximately the heating of a dense, high-Z material (solid gold) by radiation heat conduction. In this section we outline the modeling which led us to these values. The heated material is considered as a partially ionized plamsa composed of free ions (with mean charge $\overline{Z}$ and mass m_i) and electrons; the ionization energy is neglected. The plasma is assumed to be neutral with $N_e=\overline{Z}N_i$, where N_e and N_i are the number densities of electrons and ions, respectively. The plasma pressure is given by

$$p=N_ik(\overline{Z}T+T_i)=NkT(\overline{Z}+T_i/T),$$

where T and T_i are the electron and ion temperatures, and k the Boltzmann constant. We allow a temperature-dependent ionization according to

$$(\bar{Z} + T_i/T) = Z_0 T^{\vartheta},$$

where Z_0 and ϑ are constant. From this it follows that

$$pv = KT^{\delta},$$

where $K = kZ_0/m_i$ and $\delta = \vartheta + 1$. Electrons and ions have f degrees of freedom. The internal energy per unit mass is then given by

$$e = \frac{f}{2} k \frac{(\bar{Z}T + T_i)}{m_i} = \frac{pv}{\gamma - 1},$$

where $\gamma = (f + 2)/f$. As numerical values we have assumed $K = kZ_0/m_i = 7.4\times 10^3\ \mathrm{cm^2\ sec^{-2}\ {}^\circ K^{-3/2}}$ and $\delta = \vartheta + 1 = 3/2$ for gold ($m_i = 3.27\times 10^{-24}$ g). This corresponds to the relation

$$(\bar{Z} + T_i/T) = 60\sqrt{T(\mathrm{keV})}.$$

For γ we assumed $\gamma = 1.5$.

For radiation heat condution the expression for the heat flux is given by

$$S = -\frac{16}{3}\sigma T^3 l_R \frac{\partial T}{\partial x},$$

where σ is the Stefan–Boltzmann constant.

The Rosseland radiation mean free path l_R due to bound–free and free–free absorption is of the form (see Eq. 5.55 in Ref. 1)

$$l_R = A_l T^j v^{\mu'}.$$

For the multiply ionized region we choose $j = 2$ and $\mu' = 2$. For gold ($\rho_s = 19.3\ \mathrm{g\,cm^{-3}}$) one calculates $A_l = 8.6\times 10^{-16}\ \mathrm{g^2\ cm^{-5}\ {}^\circ K^{-2}}$. By comparing $S = -\kappa\partial T/\partial x$ with $\kappa = \kappa_0 T^{\nu} v^{\mu}$, one obtains

$$\kappa = \tfrac{16}{3}\sigma A_l T^{j+3} v^{\mu'}.$$

Hence, in our case we have $\nu = j + 3 = 5$, $\mu' = \mu = 2$, and $\kappa_0 = (16/3)\sigma A_l$. For the constant a and λ it follows that $a = (\kappa_0/\delta)K^{(\nu+1)/\delta} = 5.7\times 10^{-35}\ \mathrm{sec^5\ g^3\ cm^{-13}}$ and $\lambda = (\nu + 1 - \delta)/\delta = 3$.

[1] Ya. B. Zel'dovich and Yu. P. Raizer, *Physics of Shock Waves and High-temperature Hydrodynamic Phenomena* (Academic, New York, 1966).
[2] E. Buckingham, Phys. Rev. (Ser. II) **4**, 345 (1914).
[3] L. I. Sedov, *Similarity and Dimensional Methods in Mechanics* (Academic, New York, 1959).
[4] G. I. Barenblatt, *Similarity, Self-similarity, and Intermediate Asymptotics* (Consultants Bureau, New York, 1979).
[5] R. E. Marshak, Phys. Fluids **1**, 24 (1958).
[6] S. I. Anisimov, JETP Lett. **12**, 287 (1970).
[7] A. Barrero and Juan R. Sanmartín, Phys. Fluids **20**, 1155 (1977).
[8] Juan R. Sanmartín and A. Barrero, Phys. Fluids **21**, 1957 (1978).
[9] Juan R. Sanmartín and A. Barrero, Phys. Fluids **21**, 1967 (1978).
[10] J. L. Bobin, Phys. Fluids **14**, 2341 (1971).
[11] K. Nozaki and K. Nishihara, J. Phys. Soc. Jpn. **48**, 993 (1980).
[12] Iu. V. Afanas'ev, V. M. Krol', O. N. Krokhin, and I. V. Nemchinov, Appl. Math. Mech. **30**, 1218 (1967).
[13] J. Bernstein and F. J. Dyson, General Atomic Report No. GA-848, 1959; see also B. H. Armstrong, R. R. Johnston, and P. S. Kelly, J. Quant. Spectrosc. Radiat. Transfer **5**, 55 (1965).